(i) No ending after $2n$ tosses: $P = 2(pq)^n$

$$\lim_{n\to\infty} 2(pq)^n = 0 \implies P(\text{game never ends}) = 0$$

Let $P_H = P(A \text{ wins} \mid \text{first toss is H}),\ P_T = P(A \text{ wins} \mid \text{first toss is T})$

$$P_H = p + qpP_H \implies P_H = \frac{p}{1-pq}$$

$$P_T = pP_H = \frac{p^2}{1-pq}$$

$$\begin{aligned} P(A \text{ wins}) &= pP_H + qP_T \\ &= p \cdot \frac{p}{1-pq} + q \cdot \frac{p^2}{1-pq} \\ &= \frac{p^2(1+q)}{1-pq} \end{aligned}$$

(ii)
From H:

$$HH:\ p^2 \Rightarrow A \text{ wins}$$

$$T:\ q \Rightarrow P_T,\ HT:\ pq \Rightarrow P_T$$

$$P_H = p^2 + (q+pq)P_T$$

From T:

$$TT:\ q^2 \Rightarrow B \text{ wins}$$

$$H:\ p \Rightarrow P_H,\ TH:\ pq \Rightarrow P_H$$

$$P_T = (p+pq)P_H = (1-q^2)P_H$$

$$P_H = p^2 + (q+pq)(1-q^2)P_H = p^2 + (1-p^2)(1-q^2)P_H$$

$$P_H = \frac{p^2}{1-(1-p^2)(1-q^2)}$$

$$\begin{aligned} P(A \text{ wins}) &= pP_H + qP_T \\ &= pP_H + q(1-q^2)P_H \\ &= (1-q^3)P_H \\ &= \frac{p^2(1-q^3)}{1-(1-p^2)(1-q^2)} \end{aligned}$$

(iii)

$$P_H = p^{a-1} + q(1+p+\cdots+p^{a-2})P_T = p^{a-1} + (1-p^{a-1})P_T$$

$$P_T = (1-q^{b-1})P_H$$

$$P_H = p^{a-1} + (1-p^{a-1})(1-q^{b-1})P_H = \frac{p^{a-1}}{1-(1-p^{a-1})(1-q^{b-1})}$$

$$\begin{aligned} P(A \text{ wins}) &= pP_H + qP_T \\ &= pP_H + q(1-q^{b-1})P_H \\ &= (1-q^b)P_H \\ &= \frac{p^{a-1}(1-q^b)}{1-(1-p^{a-1})(1-q^{b-1})} \end{aligned}$$

When $a = b = 2$,

$$\begin{aligned} P(A \text{ wins}) &= \frac{p(1-q^2)}{1-(1-p)(1-q)} \\ &= \frac{p^2(1+q)}{1-pq} \end{aligned}$$

This solution was submitted by **Jon**

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